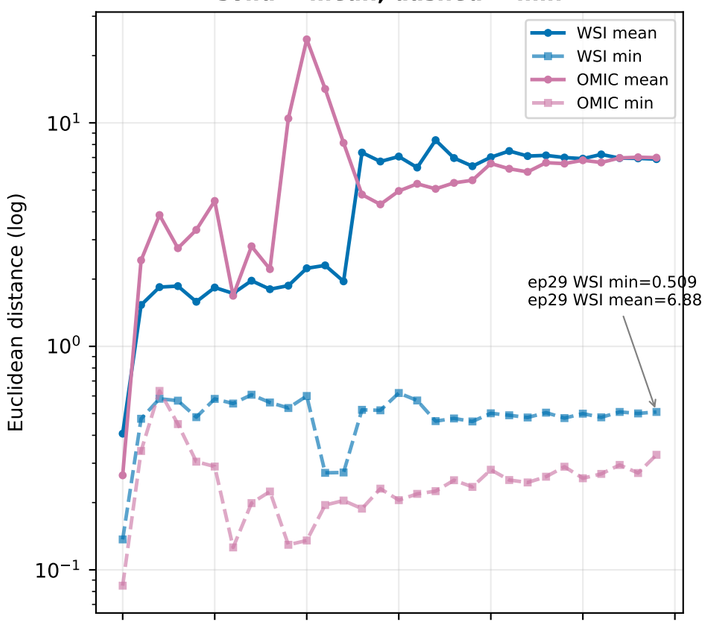
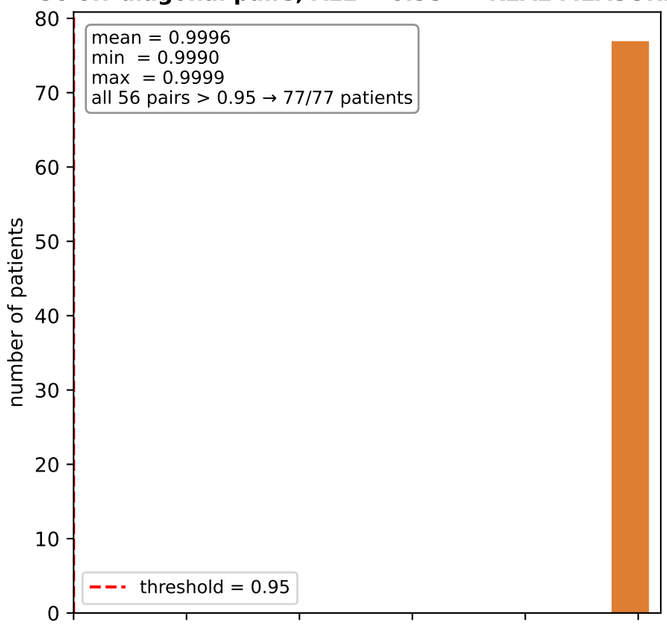
# P2 (slot redundancy): v3.13 vs SlotSPE — ALL REAL MEASUREMENTS
## v3.13 cosine NOW MEASURED (2026-10-21, 76 patients) | ⚠ single checkpoints, different protocols

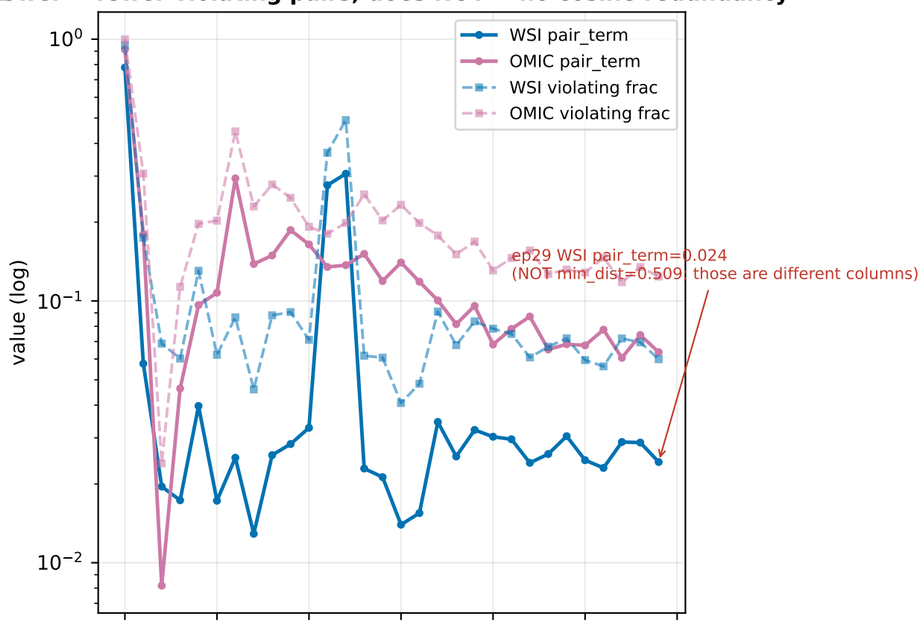
**(a) v3.13: Euclidean pair distance (BLCA fold0 seed3)**
**solid = mean; dashed = min**



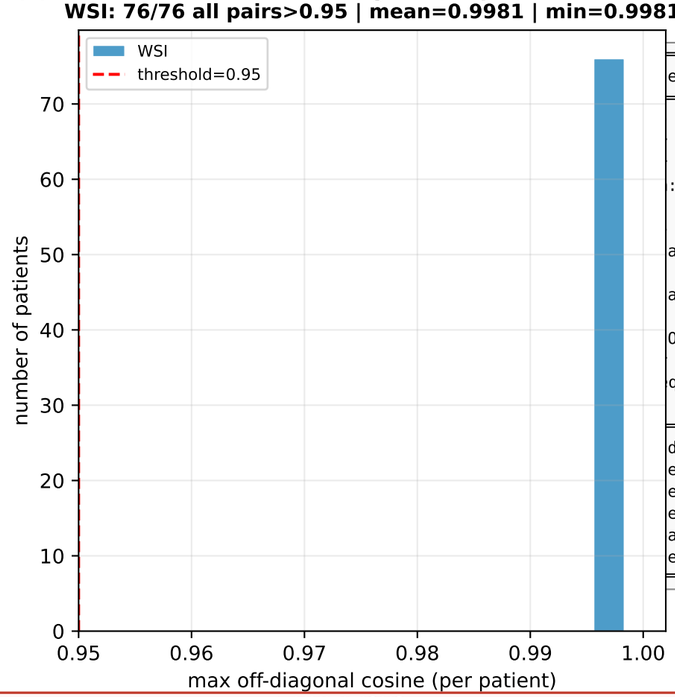

**(b) SlotSPE WSI: per-patient max cosine (77 patients)**
**56 off-diagonal pairs; ALL > 0.95 — REAL MEASURED**



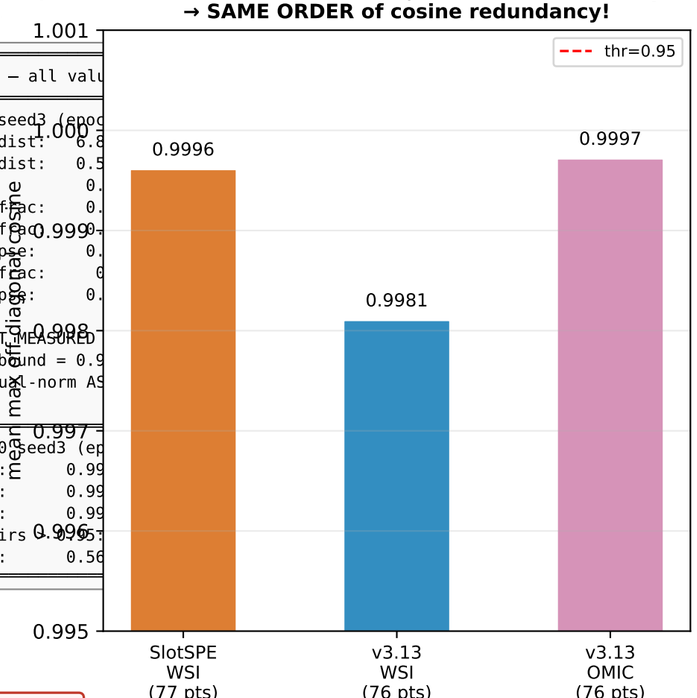

**(c) v3.13: diversity penalty (solid) + violating frac (dashed)**
**lower = fewer violating pairs; does NOT = no cosine redundancy**




**(e) v3.13 WSI: max cosine histogram (76 pts, REAL MEASURED)**
**WSI: 76/76 all pairs>0.95 | mean=0.9981 | min=0.9981**




**(f) Mean max cosine: SlotSPE vs v3.13 (both REAL MEASURED)**
**SlotSPE WSI: 77/77 all pairs>thr | v3.13 WSI: 76/76 all pairs>thr**
**→ SAME ORDER of cosine redundancy!**




*⚠ LIMITATIONS:*
*(a)(c) v3.13: BLCA fold0 seed3 best checkpoint (not 5-fold mean).*
*(b) SlotSPE: BLCA fold0 seed3 epoch3 (not 5-fold mean).*
*v3.13 cosine NOW MEASURED (2026-10-21) from real checkpoint forward pass.*
*v3.13 Euclidean dist improves (0→6.88) but cosine stays near 1.0 — same order as SlotSPE.*
*Source: scripts/measure_v313_slot_cosine.py → v313_slot_cosine_blca_fold0_measured.json*